

“Dongba Script Character Construction Space”: VR Science-Based Interactive Experience of Pictographs in Intangible Cultural Heritage

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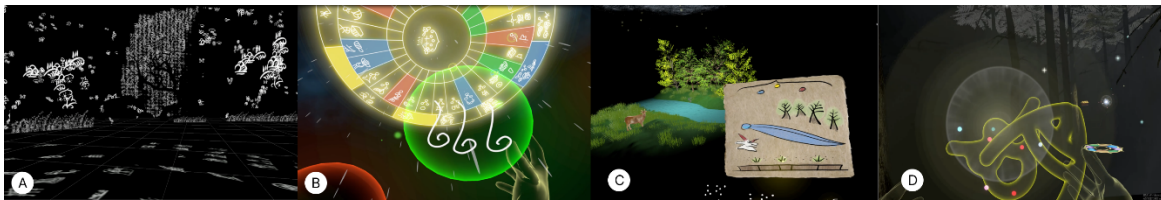


Figure 1: The initial scenario and three composition word showroom: (a) Initial scenario (b) Pictographs. (c) Simple ideographs. (d) Compound ideographs.

ABSTRACT

The "Dongba Script Character Construction Space" is an interactive science VR experience based on the Dongba script of the Naxi people in Yunnan Province, China, which provides an immersive virtual reality experience based on the traditional Yiyin script theory. [Selmanović et al., 2020] Specifically, this work combines the character construction characteristics of "pictogram", "simple ideographs" and "compound ideographs" to explore the meaning and visual symbolic characteristics of Naxi pictograms through three interesting forms of interaction, and provides more possibilities for the inheritance and development of Dongba in the future by using various forms of interaction.

CCS CONCEPTS

• **Human-centered computing**; • **Virtual reality**; • **Arts and humanities**;

KEYWORDS

Intangible cultural heritage, Virtual reality, Interaction design

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1 INTRODUCTION

The Dongba script is called the world's only surviving hieroglyphic script, and is known as a "living fossil" of writing. Its basic feature is to use graphics to indirectly represent the meaning of words. [Zhou, 2019] As the number of inheritors decreases year by year, the severely damaged ancient Dongba script, the difficulty of cultivating literate people to read the Dongba script, and the challenge of passing on and popularizing the Dongba language. Based on the traditional "six writings" theory, we designed three virtual exhibition halls representing the characteristics of the Dongba script, combining interactive content with interactive forms to bring better learning and memorization of the script for the participants, and to fully demonstrate the unique charm of the Dongba script as an intangible cultural heritage.

2 IMPLEMENTATION

The "Dongba Script Character Construction Space" consists of an initial scene and three character exhibition halls. We designed three interactive forms based on the features of three types of character construction methods, namely "pictographs", "simple ideographs" and "compound ideographs". Three character experience exhibition halls were formed for an understanding of the characteristics of these character construction methods as well as the meaning of Dongba script. Three forms of interaction are nonlinear, and three interactive exhibition halls are arranged in three different directions from the initial scene. Audience can wear VR headsets and controllers to randomly select the interactive exhibition hall to experience (see Figure 1).

3 VIRTUAL SCENARIOS

3.1 Initial Scenario

This scene uses traditional Chinese ink colors and simulates natural landscapes such as blooming flowers, waterfalls, rivers, grasslands, boulders, and trees using Dongba script. Three different flowing text rivers guide visitors to three different Dongba script spaces. The dynamic, interlaced text is filled with vivid natural forms, creating a flowing Dongba script natural system that corresponds to the Naxi people's love for nature and their reverence for it.

3.2 Pictographs

The ideograms in Dongba script refer to using simple lines to depict the shape and features of real objects, or indirectly represent the meaning of language through graphical forms. For example, the Dongba script for "rain" looks like raindrops falling down. In north-west Yunnan, where the Naxi live, the rainy season means that everything grows, so colorful flowers grow on the grass while it rains. In this scene, the Twenty-Eight Constellations chart is placed above, and there are constantly text balls falling from it, guiding users to interact by looking up. Users learn more about pictograms by catching falling spheres to get feedback on scenes related to pictograms.

3.3 Simple Ideographs

The simple ideographs in the Dongba script are created by adding "scattered points" and "curves" to things to represent abstract concepts of action, state, property and behaviour. Nearby is a painting of Dongba characters, and in the distance, there is a scene that matches the painting. There are "scatter points" and "curves" under the Dongba painting that can be picked up. By randomly picking up "scattered points" or "curves" and placing it into the characters in the Dongba painting, the user creates a new character and triggers a change in the distant model that mirrors the character, allowing the user to repeatedly trigger a comparison between the two states.

3.4 Compound Ideographs

The compound ideographs in the Dongba character are formed by combining two or more pictographs to represent a new word. [Song et al., 2010] Participants gain a deeper understanding of the characteristics and textual meanings of the characters by using two forms of interaction: pulling apart and merging. The most important feature of compound ideographs is the combination and separation of the character components. These two interactive methods fully embody the uniqueness of associative characters. Understanding the logical transformation from a single character to a composite character helps to better understand the traditional cultural psychology of the Naxi ethnic group.

4 SUMMARY

Most viewers have expressed that Dongba Script Character Construction Space exhibits unique ethnic features, and the three interactive modes are both scientifically interesting. Through this entertaining interaction, they have gained a deeper understanding of the structural characteristics of Dongba script. The audience believes that the part of "Compound ideographs" is more unique, and

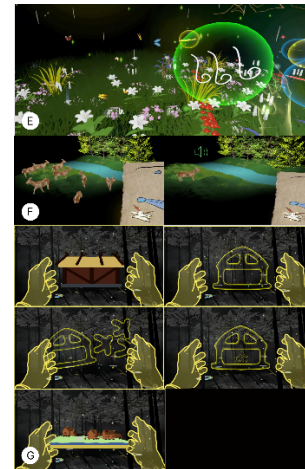


Figure 2: (e) The "rain" in the pictographs scene. (f) Comparison of "scattered points" and "curves". (g) Compound ideographs interaction process.

the continuous interaction between "pulling apart" and "merging" enables them to learn in entertainment. This work combines character construction with interactive behavior and utilizes a scientific and fun interactive experience to create the Dongba spatial character construction, providing initial exploration of digital inheritance and popularization of Dongba script in the future (see Figure 2).

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